Speech-To-Text and NLP: Comparing De Gasperi and Meloni's Rhetoric

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Comparison between Alcide De Gasperi and Giorgia Meloni rhetoric. After building a corpus of public speeches transcribed whit Whisper from Meloni's videos on Youtube, I compare them with a corpus of speeches from Italy's first Prime Minister Alcide De Gasperi. TODO

CCS Concepts: • **Computing methodologies → Natural language processing**; **Speech recognition**; • **Social and professional topics → Hate speech**; **Political speech**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [13] showed how nearly all 20^{th} century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [2]: leaders depend on the verbal power to persuade people [4].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [6] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57,000 times between May 2009 and January 2021 alone [10], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [17].

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover,

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until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [8, 12].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this mini-project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as sentiment analysis and hate speech detection.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [5] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [7, 11]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [1, 9], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [14].

Opposite results can be seen in Belcastro et al.'s [3] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [15] managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

That being said, X is just one channel among many used to connect with the voters: official messages to the nation, interviews, and public speeches are still powerful tools, and are often made publicly available in video format by the politicians themselves on their Facebook page and Youtube channel.

3 PROJECT GOAL

Let's now define the goal of the mini-project.

I want to compare the rhetoric between the first and the current[1] Italian Prime Minister, namely Alcide De Gasperi and Giorgia Meloni. The rhetoric will be compared using natural language processing techniques, such as measuring textual complexity and hate speech classification. Only public speeches will be taken, thus leaving out all parliamentary speeches.

I also want to produce a proof-of-concept for a speech-to-text pipeline that, given a set of URLs, each of which a Youtube's permalink, transcribe the video into text.

[1]At the moment of writing: May 2026.

The first step will be gathering two corpora of public speeches, one for each PM. In the case of Giorgia Meloni, I want to compile the corpus by transcribing video interviews and public talks available on her official Youtube channel.

Then, after having gathered the corpora into suitable machine-readable data (such as csv), I will pre-process them with procedures such as tokenization, stemming and lemmatization. Lastly, I will compute word frequency and text complexity measures, as well as hate speech classification and sentiment analysis.

4 BUILDING THE CORPUS

4.1 Alcide De Gasperi's Corpus

I would like to thank Sara Tonelli for providing me with the De Gasperi's Corpus [16], a collection of Alcide De Gasperi's public documents with gold and silver annotation.

The corpus is a collection of 2,762 documents issued between 1901 and 1954, formatted into XML files which include metadata that covers not only the title, the date and the place of publication, but also key-concepts automatically extracted from each text (with the corresponding relevance score) and genre labels manually assigned by domain experts. Furthermore, the release includes silver annotation for lemma, part of speech, person names and place names with associated coordinates in a CoNLL-like format.

An example of the XML formatting can be seen at listing 1. Thanks to the *<genres>* tag, I was able to extract 474 public speeches, each of which with a location, a precise date and a list of keywords.

Listing 1. An example of how a single speech in De Gasperi's Corpus is formatted in XML

```xml
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<document id="I.doc_7">
   <publication_date>1901-11-26</publication_date>
   <publication_place>
      <place_name>Trento</place_name>
      <place_latitude>46.0664228</place_latitude>
      <place_longitude>11.1257601</place_longitude>
   </publication_place>
   <keywords>
      <keyword score="39.16">first keyword</keyword>
      ...
      <keyword score="5.59">last keyword</keyword>
   </keywords>
   <genres>
      <genre>Speech Type</genre>
   </genres>
   <text>The speech itself if stored between these brackets</text>
</document>
```

4.2 Giorgia Meloni's Corpus

My objective was to build Meloni's corpus by transcribing videos of her public speeches, such as talks and interviews. Fortunately, there is an unofficial Youtube channel (that I reached from the Prime Minister's official website) which aggregates more than four thousand videos of her public appearances. The channel is called "Giorgia Meloni News"[2].

[2]Giorgia Meloni News: https://www.youtube.com/@GiorgiaMeloniTv

Given a Youtube URL, I can use Python's library *yt-dlp* to retrieve the video's metadata and download its audio content in mp3 format, as shown in listing 2. I had to manually pass the cookies taken from my web-browser, and I used some extra commands to prevent Youtube to block the requests due to suspicious activity.

Then, the mp3 file just retrieved gets injected into OpenAI's Whisper library, which uses an encoder-decoder Transformer to transcribe it into text, as shown in listing 3. The model I used is the *"medium"*[3], which at 5 GB fits within the total 6 GB of VRAM available to my RTX 4050 GPU. The model was instructed to predict Italian.

The difference in precision between the models "tiny" (which requires less than 1 GB of VRAM) and "medium" is quite high, as can be seen in the following pieces of text that have been generated from the same audio file:

- **Model tiny:** "iamo con se è beruto fuori da questa rio neone sotto il profiro tecnico per quanto riguarda la franha e come il governo un tino era ad essere vicino alla popolazione di Nishenia."
- **Model medium:** "Diciamo cosa è venuto fuori da questa riunione sotto il profilo tecnico per quanto riguarda la frana e come il governo continuerà ad essere vicino alla popolazione diniscemeca."

The last step of the pipeline saves the transcription into a csv file, along with the extracted metadata. Each file has the following fields: *politician* ("meloni" in this case), *historical_date* (the upload date of the video), *location* and *tags* (extracted from the metadata if available, empty strings otherwise), *description* and *title* of the video, *url* which stores the permalink of the video itself, and lastly *text* which holds the whole transcription.

Each video is processed immediately after being retrieved, and until it has been saved in a csv file the program does not try to fetch the next one. This is done for two reasons: first, to prevent *yt-dlp* from sending requests too close to each other, which could result in Youtube denying them due to suspicious activity. Second, even if the pipeline halts for any reason, the videos processed up to that point will not be lost.

Listing 2. Downloading audio and metadata from a Youtube video

```
result = subprocess.run([
    "yt-dlp",
    "--print-json",
    "-x",                               # download audio only
    "--audio-format", "mp3",
    "--cookies", "yt_cookies.txt",      # manual cookies, extract with browser extension
    "--sleep-requests", "2",            # Sleep 2s between requests
    "--sleep-interval", "5",            # Sleep 5s between downloads
    "--max-sleep-interval", "15",       # Randomize up to 15s
    "--limit-rate", "5M",               # Throttle to 5MB/s (mimics streaming)
    "-o", audio_file.replace('.mp3', ''),
    video_url
], capture_output=True, text=True, check=True)
```

Listing 3. Transcribe the audio file into text using Whisper

```
subprocess.run([
    "whisper",
    "--language",         lang_code,    # "Italian" chosen
    "--word_timestamps", "True",
    "--model",            model_name,   # model "medium" chosen
    "--output_dir",       output_dir,
    "--device",           "cuda",       # ensures the model gets loaded in GPU
    audio_file
```

[3] Whisper model sizes available are tiny, base, medium and large.

```
], check=True)
```

5 TEXTUAL ANALYSIS

All the code used to compute the results shown in this section can be seen in the jupiter notebook mhetacc/cross-dem/notebooks/text_analysis.ipbynb.

5.1 Token Count and Textual Complexity

Overall the collected corpus amount of 473 speeches from De Gasperi, and 72 from Meloni, with a total token count of 609, 142 for De Gasperi and 120, 116 for Meloni. On average, Meloni's corpus have 1668 tokens per speech, while De Gasperi's 1287 tokens per speech. Results in table 1.

I then measured the textual lexical diversity using *MTLD*, which I preferred over the more common *TTR* (Type-Token Ratio) since MTLD is agnostic on the corpus size. I used the *LexicalRichness* library[4] to compute the MTLD for each speech, and then both the average and the median MTLD for both PMs. Results can be seen in table 2.

To better visualize the difference of textual diversity I plotted the MTLD values of 70 different speeches of Meloni and De Gasperi against each other. The plot can be seen in figure 1.

Table 1. Total tokens and average tokens per speech for De Gasperi and Meloni.

Politician	Total Tokens	Speech Count	Avg Tokens/Speech
De Gasperi	609,142	473	1,287.83
Meloni	120,116	72	1,668.28

Table 2. MTLD metrics for De Gasperi and Meloni.

Politician	Avg MTLD	Median MTLD
De Gasperi	141.830	137.642
Meloni	106.745	104.918

5.2 Vocabulary Differences

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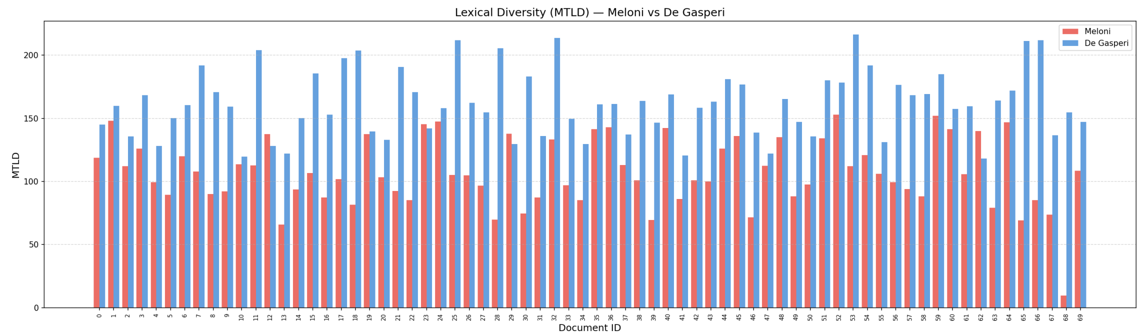

Fig. 1. Raw MTLD values for 70 speeches of Meloni and De Gasperi

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